



**TAKSHASHILA
UNIVERSITY**

AI-POWERED MULTIMODAL E-COMMERCE SEARCH SYSTEM USING FAISS AND HYBRID RETRIEVAL

By

MANGIPUDI SANYASI RAO

Enroll. No. TU6243204211024

MCA(DS)-II

Under the Guidance of

Ms.V. KIRUTHIGA MCA., M.Phil., Ph.D.,

Assistant Professor,

School of Computer Science

INTRODUCTION



E-commerce platforms contain a **large and diverse collection of products**



Users often face difficulty in finding relevant products quickly



Traditional search systems rely mainly on **keyword matching techniques**



These systems fail to understand the **actual intent behind user queries**

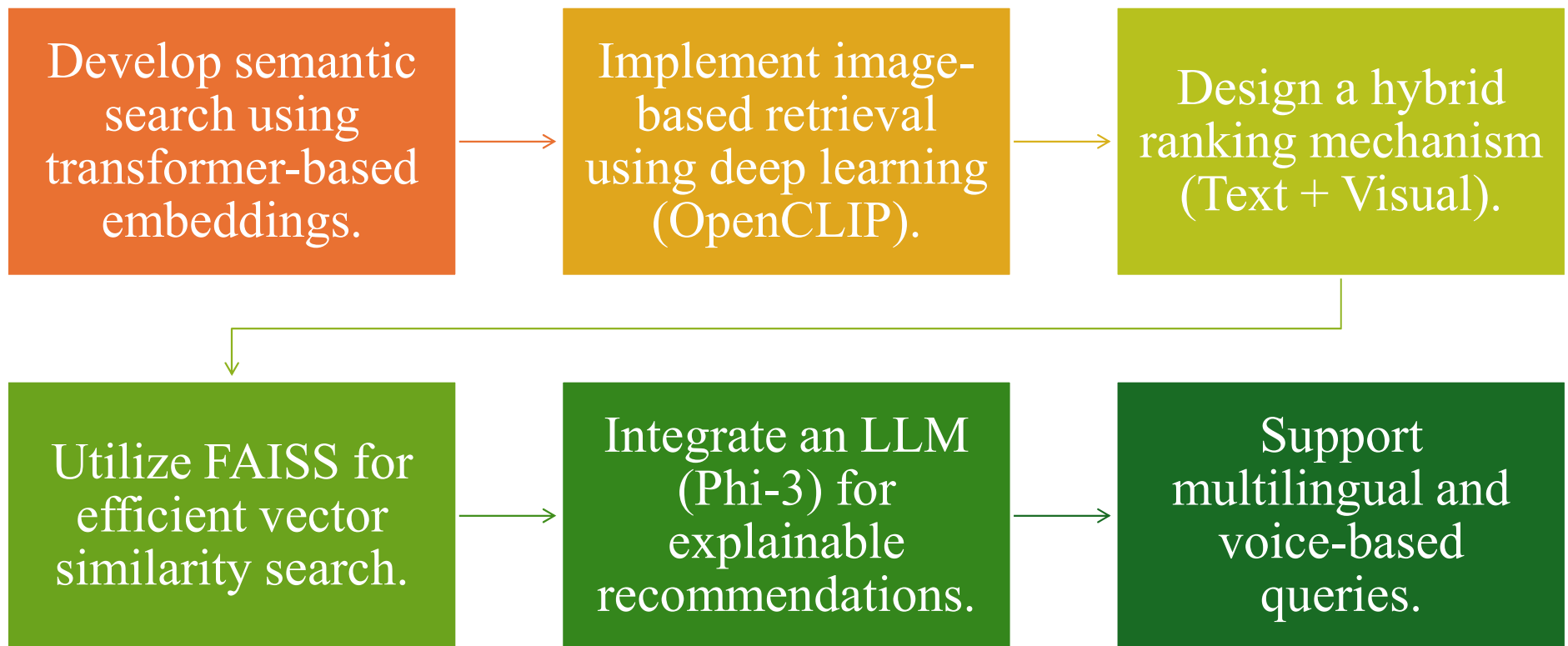


This leads to irrelevant results and poor user satisfaction



AI-based systems provide **semantic understanding and intelligent recommendations**

OBJECTIVE OF THE PROJECT



SCOPE OF THE PROJECT

Included:

Processing structured/unstructured product data.

Generating embeddings for text and images.

Vector-based similarity search using FAISS.

Hybrid ranking & LLM explanation generation.

Responsive web-based UI with chatbot.



Boundary: Designed for scalability but tested on a moderate-sized dataset.

LITERATURE SURVEY & RESEARCH GAP

Author(s)	Year	Contribution	Limitation / Gap
Reimers & Gurevych (Sentence-BERT)	2019	Semantic sentence embeddings for similarity search	Text-only; no visual understanding
Radford et al. (CLIP)	2021	Multimodal (text + image) shared embedding space	Retrieval not optimized for e-commerce scale
Johnson et al. (FAISS)	2017	Efficient similarity search for dense vectors	No ranking or explanation generation
Rusum & Anasuri (Vector Databases)	2024	Real-time search & RAG pipelines	Lacks hybrid retrieval + LLM explainability
Ali, O. (RAG)	2025	Retrieval-augmented generation for databases	No integration with multimodal e-commerce search

RESEARCH GAP IDENTIFIED

GAP	DESCRIPTION
1. No Unified Multimodal System	Existing systems handle text OR image, not both together
2. No Hybrid Ranking	Semantic search lacks keyword boosting; exact matches ignored
3. No Explainability	Search results are shown without reasoning or justification
4. No RAG for E-commerce	LLMs are not integrated with product retrieval for explanations

HOW PROPOSED SYSTEM ADDRESSES THESE GAPS

Research Gap	Proposed System Solution
Multimodal missing	Text + Image search using Sentence Transformers + OpenCLIP
Poor ranking	Hybrid Ranking (Semantic + Keyword + Filters)
No explainability	LLM (Phi-3) generates factual product explanations
No RAG integration	RAG-inspired pipeline retrieves products → LLM explains

TECHNOLOGIES USED

Programming: Python, FastAPI (Backend), HTML/CSS/JS (Frontend).

NLP: Sentence Transformers (BAAI/bge-small-en-v1.5).

Computer Vision: OpenCLIP (ViT-B/32).

Vector Search: FAISS (Facebook AI Similarity Search).

LLM: Phi-3 (via Ollama).

Multilingual: Langdetect, Deep Translator.

EXISTING SYSTEM (LIMITATIONS)



Keyword Dependency: Fails on synonyms (e.g., "affordable" vs. "cheap").



No Visual Search: Cannot find products by image similarity.



Poor Ranking: Results lack semantic relevance.



No Explainability: Users don't know *why* a product was recommended.

PROPOSED SYSTEM METHODOLOGY (ARCHITECTURE OVERVIEW)



Input Layer: Text, Voice, Image, Multilingual.



Processing Layer: Sentence Transformers / OpenCLIP -> FAISS Index.



Ranking Layer: Hybrid Scoring (Semantic + Keyword + Filters).



Generation Layer: LLM (Phi-3) for Explanations.



Output Layer: Ranked products with descriptions.

DATA DESIGN & DATASET ENGINEERING



Source: Merged multiple public datasets (Fashion, Electronics).



Size: Unified structured format.



Key Attributes: Title, Description, Price, Category, Image Path.



Engineered Feature: search_text (Combined title+desc+tags for better embeddings).



Preprocessing: Cleaning, deduplication, standardization.

EMBEDDING GENERATION (TEXT & IMAGE)

Text Embeddings:

- Model: BAAI/bge-small-en-v1.5.
- Output: 384/768-dim vectors capturing semantic meaning.

Image Embeddings:

- Model: OpenCLIP (ViT-B/32).
- Output: Visual feature vectors (color, shape, texture).

Optimization: GPU acceleration reduced time from 5-6 hours to minutes.

FAISS Index Construction



Why FAISS? Enables Approximate Nearest Neighbor (ANN) search for millions of vectors in milliseconds.



Implementation:

Index 1: Text FAISS Index (Semantic search).

Index 2: Image FAISS Index (Visual similarity).



Storage: Binary (.bin) files for persistence.



Mapping: Index positions mapped to Product IDs

HYBRID RETRIEVAL MECHANISM

Limitation of Single Method: Semantic search might miss exact keywords.



Formula:
$$\text{Final Score} = (w1 * \text{Semantic Similarity}) + (w2 * \text{Keyword Boost})$$



Process:

Retrieve candidates via FAISS.

Re-rank using hybrid scoring.

Apply attribute filters (Price, Category).



Outcome: Highly relevant and precisely matched results.

LLM INTEGRATION FOR EXPLAINABILITY (RAG)



Problem: LLMs hallucinate.



Solution: Retrieval-Augmented Generation (RAG) approach.



Workflow:

1. Retrieve product data (FAISS).
2. Feed data as context to LLM (Phi-3).
3. Generate factual, concise explanation.



Example: "This product is recommended because it matches your query for 'red shoes' and has a 4.8 rating."

MULTILINGUAL & VOICE PROCESSING

Multilingual:

- Detect language (Langdetect).
- Translate to English (GoogleTranslator).
- Process query normally.
- *Supported:* Hindi, Tamil, Telugu, etc.

Voice Search:

- Browser Speech Recognition API.
- Convert speech -> text -> search.

IMPLEMENTATION (BACKEND - FASTAPI)

Framework: FastAPI (Async, high-performance).

Key Endpoints:

- GET /search?query= -> Text search.
- POST /image-search -> Image upload search.
- GET /chat?msg= -> Chatbot response.

Middleware: CORS for frontend-backend communication.

Response: JSON containing products, scores, and explanations.

IMPLEMENTATION (FRONTEND - UI/UX)

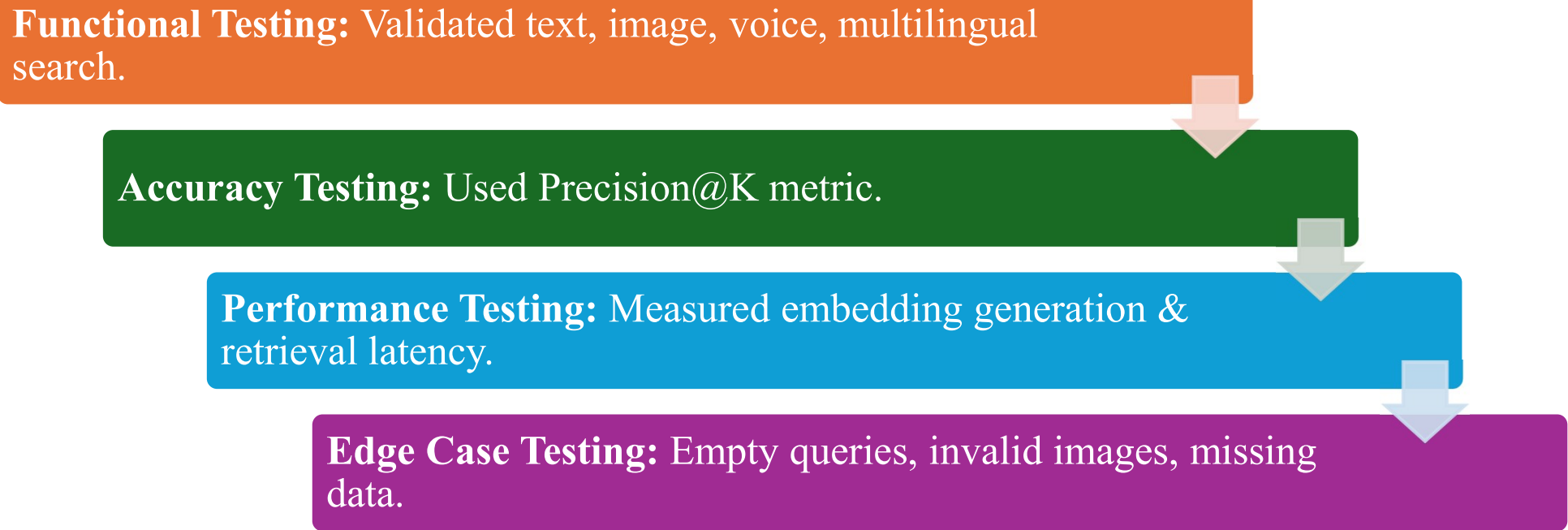
Technologies: HTML5, CSS3, JavaScript.

Key Features:

- Semantic search bar.
- Image upload widget.
- Voice input button.
- Dark mode toggle.
- Shopping cart.
- Chatbot assistant.

SYSTEM TESTING APPROACH

Functional Testing: Validated text, image, voice, multilingual search.



```
graph TD; A[Functional Testing: Validated text, image, voice, multilingual search.] --> B[Accuracy Testing: Used Precision@K metric.]; B --> C[Performance Testing: Measured embedding generation & retrieval latency.]; C --> D[Edge Case Testing: Empty queries, invalid images, missing data.];
```

Accuracy Testing: Used Precision@K metric.

Performance Testing: Measured embedding generation & retrieval latency.

Edge Case Testing: Empty queries, invalid images, missing data.

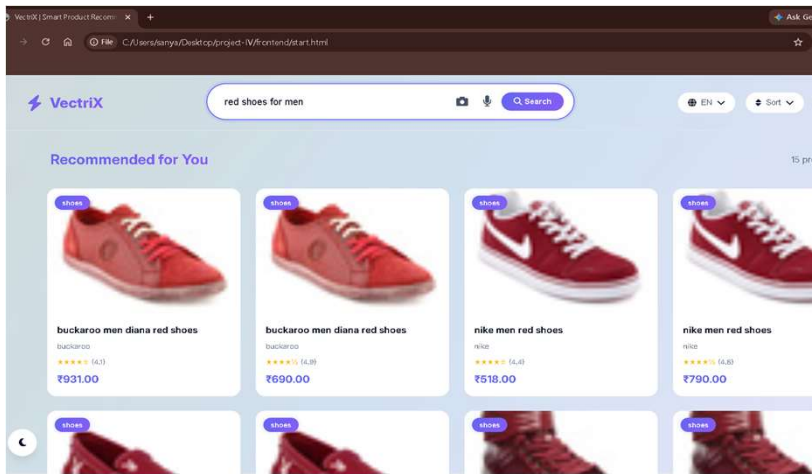
TEST RESULTS (PRECISION@K)

Query	K Value	Relevant Results	Precision@K
"red shoes"	5	4	0.8
"wireless headphones"	5	4	0.8
"sports shoes"	5	5	1.0
"black jacket"	5	3	0.6

Analysis: High accuracy for specific queries; scope to improve ambiguous terms.

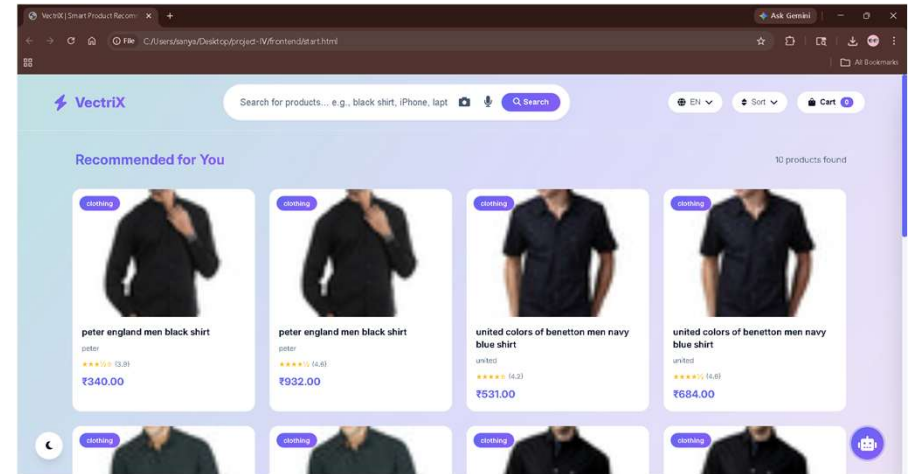
RESULTS - TEXT & IMAGE SEARCH

Text Search



Query "red shoes for men" returns semantically relevant red footwear (not just exact "red").

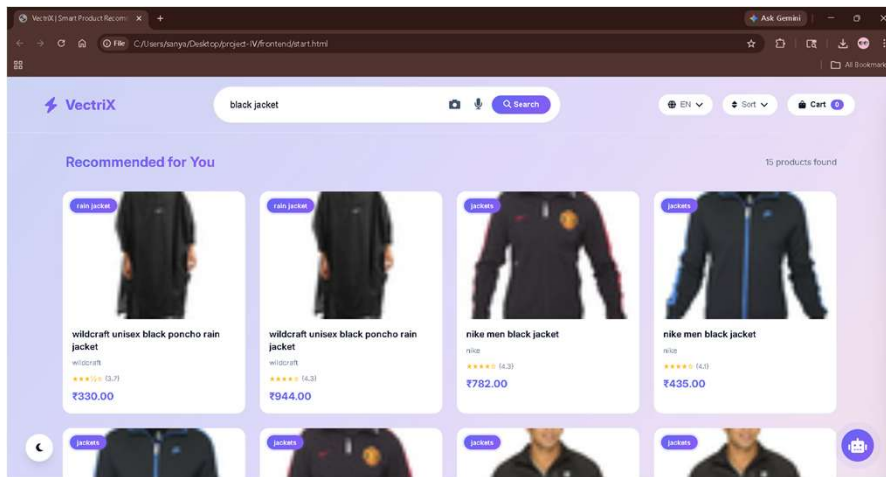
Image Search



Uploading a Black shirt image returns visually similar shirts based on shape/color.

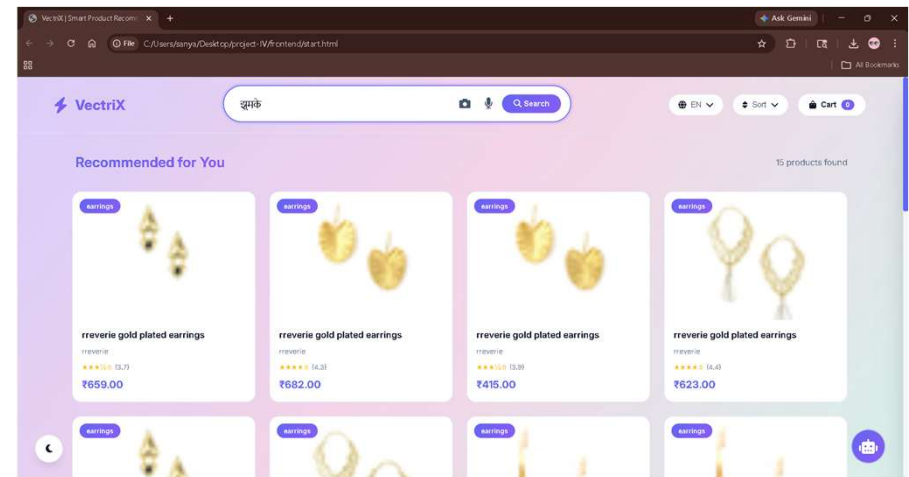
RESULTS - MULTILINGUAL & VOICE

Voice



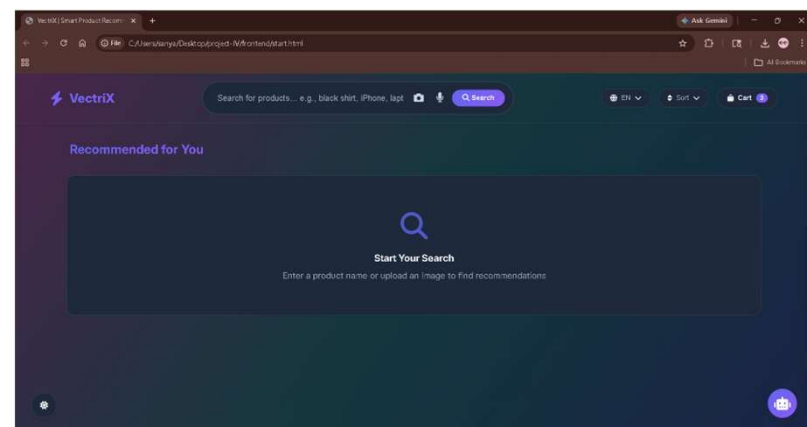
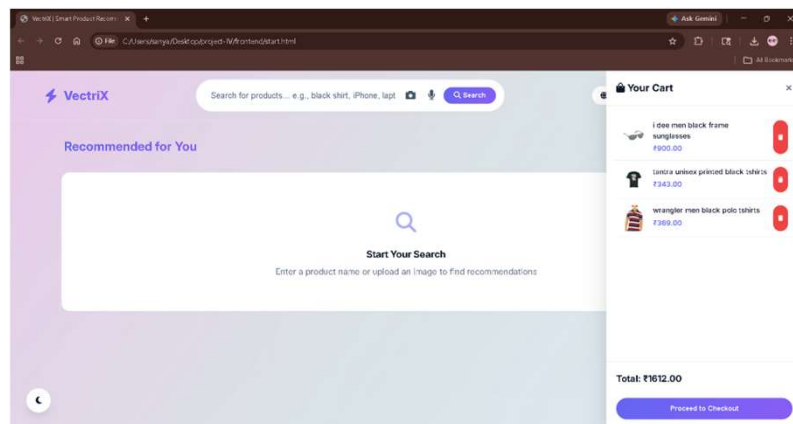
Click mic -> Speak "black jacket" -> Text fills -> Search executes.

Multilingual



Input "झुमके" (Hindi) -> Detected -> Translated -> Returns earrings.

RESULTS - UI FEATURES



- The **Shopping Cart** and **Dark Mode** features significantly enhance the user experience by prioritizing functionality and comfort.

RESULTS - CHATBOT (RAG)



Function: Integrated shopping assistant.



Query: "What should I check before buying a laptop?"



Process: FAISS retrieves laptop products -> LLM generates context-aware buying advice.



Outcome: Factual, helpful responses (not generic AI).

DISCUSSION OF RESULTS

Strengths:

- Semantic understanding significantly outperforms keyword search.
- FAISS enables real-time retrieval (<2 seconds).
- Multimodal input (text/image/voice) provides flexibility.

Challenges:

- Image retrieval accuracy varies with image quality.
- Precision drops for ambiguous queries (e.g., "black jacket").

CONCLUSION

Successfully built a **Multimodal AI E-commerce Search System**.

Integrated **NLP, CV, FAISS, and LLM** into a unified architecture.

Achieved **high Precision@K** (up to 1.0) using hybrid retrieval.

Enabled **explainable AI** via RAG-based LLM explanations.

Provided a **production-ready UI** with modern features (dark mode, voice, multilingual).

LIMITATIONS OF THE SYSTEM

Dataset Dependency: Performance tied to data quality.

LLM Capacity: Phi-3 is lightweight; less detailed than larger models.

No Personalization: Does not use user history or preferences.

Scalability: FAISS is efficient but distributed databases (Milvus) would be better for billion-scale.

FUTURE SCOPE



Advanced LLMs: Integrate LLaMA 3 / GPT-4o for deeper reasoning.



Real-time Recommendations: Add collaborative filtering for personalization.



Vector Databases: Migrate to Pinecone/Milvus for cloud scaling.



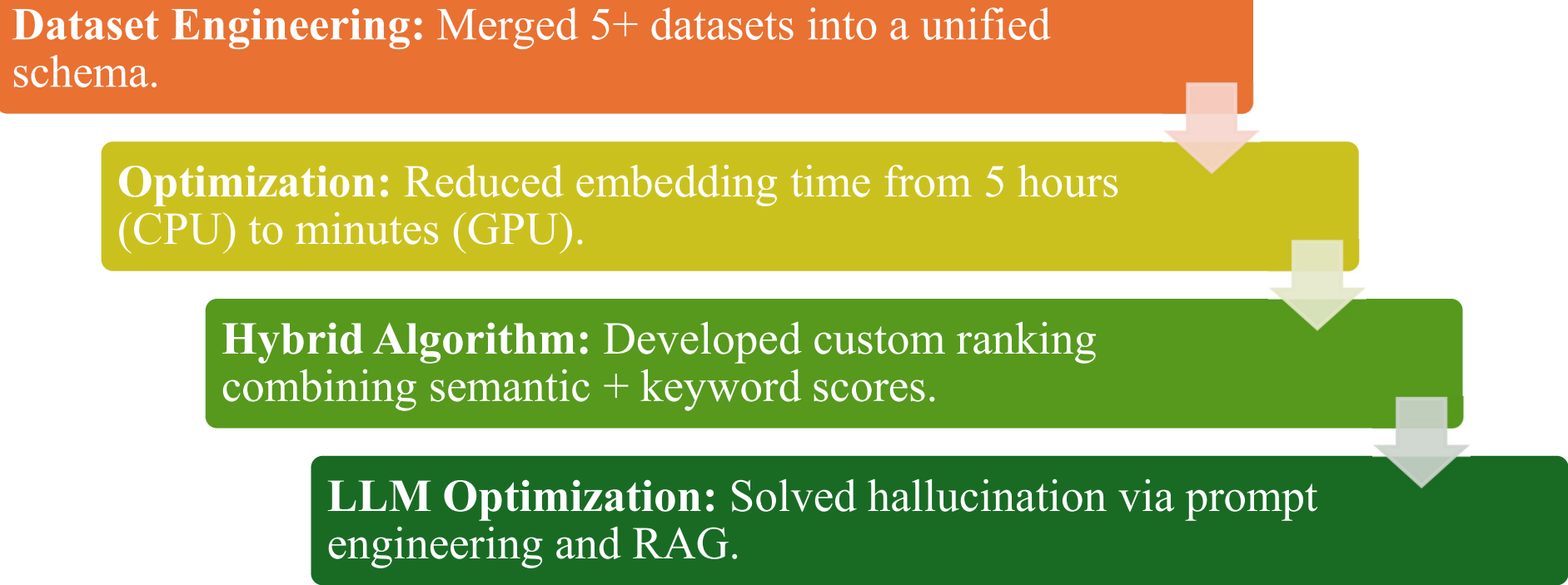
Multimodal Models: Use Gemini-style models for joint text-image understanding.



Mobile App: Deploy as React Native app with camera search.

KEY ACHIEVEMENTS (RECAP)

Dataset Engineering: Merged 5+ datasets into a unified schema.



```
graph TD; A[Dataset Engineering: Merged 5+ datasets into a unified schema.] --> B[Optimization: Reduced embedding time from 5 hours (CPU) to minutes (GPU).]; B --> C[Hybrid Algorithm: Developed custom ranking combining semantic + keyword scores.]; C --> D[LLM Optimization: Solved hallucination via prompt engineering and RAG.];
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REFERENCES

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THANK YOU

Project Guide: Ms.V. KIRUTHIGA MCA., M.Phil., Ph.D.

Student: MANGIPUDI SANYASI RAO
(TU6243204211024)

School: Computer Science, Takshashila University

